Module 4: Sampling and Estimation

Four lessons | From sample selection to uncertainty

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Learning sequence

Complete Module 3 first. We distinguish individual outcomes from sample statistics, develop standard errors, then use them to construct intervals and plan precision.

Lesson	Practice
1. Sampling and Variability	1. Practice
2. Sampling Distributions and Standard Errors	2. Practice
3. Confidence Intervals for Means	3. Practice
4. Proportions and Sample-size Planning	4. Practice

Finish the integrated review. Instructor answers are kept separately.

Cases, readings, and evidence

The main textbook is Anderson et al. (2020), 8th edition. Selected Chapter 7 material develops sampling and sampling distributions; selected Chapter 8 material develops mean and proportion intervals and planning.

- EAI employees: population summaries p.308; sample summaries p.316; finite correction and CLT pp.323–324.
- Lloyd's spending: known-SD inputs p.357.
- Household credit-card balances: rounded sample summaries and t interval pp.367–368.
- Women golfers: survey counts and normal interval p.378; planning pp.380–381.

We use stated book parameters and summaries rather than introducing unverified observation files. Artificial simulations and original planning extensions are labeled. No new workbook-to-CSV match is claimed.

Next step

Module 5: Hypothesis testing and group comparisons follows interval estimation; its existing lesson remains retained material pending expansion.

Return to course home.